WEEK 4 OVERVIEW

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CEVE 543

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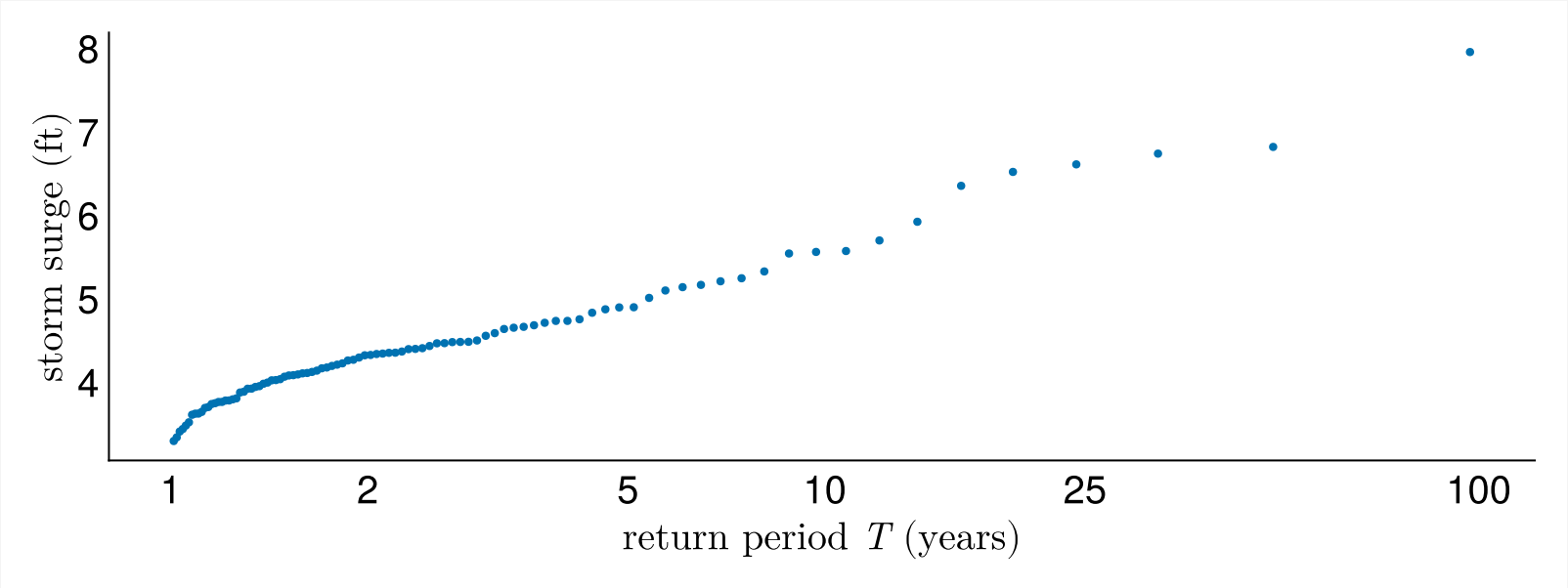


Module 2: univariate extremes

HOW DO WE CHARACTERIZE A SINGLE HAZARD AT A SINGLE LOCATION?

How do we characterize the probability distribution of a single hydroclimate hazard at a single location, and what are the associated uncertainties?

WE HAVE ALREADY WORKED WITH ANNUAL MAXIMA



Sewells Point annual maximum storm surge, 1928-2023, at Weibull plotting positions. The largest observation gets a return period of about 97 years. What is the 200-year level? The 500-year level?

EXTREME VALUE THEORY GIVES US A PRINCIPLED ANSWER

- Extrapolating beyond the data requires a model
- We would like a justification beyond “it fits”

Extreme value theory to the rescue: a limit theorem says that block maxima converge to a specific family of distributions (the GEV).

WHAT THIS WEEK COVERS

- **Monday:** the GEV distribution and how to estimate its parameters
- **Wednesday:** fitting a GEV in Julia, diagnostics, and Bayesian inference
- **Friday:** lab 4, writing your own model in Turing and testing its sensitivity